

Question	Marking Guidance	Mark	Comments
10.1	- TEM use electrons and optical use light; - TEM allows a greater <u>resolution</u>; (So with TEM) smaller <u>organelles/named cell structure</u> can be observed OR greater detail in <u>organelles/named cell structure</u> can be observed; - TEM view only dead/dehydrated specimens and optical (can) view live specimens; - TEM does not show colour and optical (can); - TEM requires thinner <u>specimens</u>; - TEM requires a <u>more</u> complex/time consuming preparation; - TEM focuses using magnets <u>and</u> optical uses (glass) lenses;	6 max	3. 'clearer' is not equivalent to 'detail' 4. Accept '<u>Only</u> optical can view live specimens' 5. Accept '<u>Only</u> optical can show colour' 7. Accept 'TEM requires a more difficult preparation' Ignore references to artefacts
10.2	W has 4 cells/nuclei since it is at the (end of) 2nd division (of meiosis); Z has 2 cells/nuclei since it is at the (end of) 1st division (of meiosis); W shows haploid cells/cells containing n chromosomes; (Cells in) W contain half the (mass of) DNA of (Cells in) Z; OR (between Z and W) chromatids have separated; OR In Z homologous chromosomes have separated;	4	Ignore 'Z shows diploid cells/contains 2n chromosomes' 4. Accept 'W contains half the amount of DNA of Z'

10.3	1. Use random sample of seeds (from each population); 2. Use (large enough) sample to be representative of whole population; 3. Indication of what size was measured eg mass; 4. Calculate a <u>mean</u> and standard deviation (for each population); 5. Use the (Student's) t-test; 6. Analyse whether there is a significant difference between (the means of) the two populations;	5 max	1. Accept described, suitable method of random sampling. 1. Reject description of inappropriate method of random sampling (eg random coordinates in the field/use of quadrats) 2. Accept 'running mean does not change' 2. For representative accept 'reliable, reproducible, repeatable' OR a mean close to the true value. 5. Accept 'Use 95% confidence limits' 6. Reject unqualified references to results being significant
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